

Nishant Kumar

PhD Researcher — Bioinformatics — Machine Learning
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Research Interest

Protein bioinformatics, Machine learning for peptides/proteins, Epitope prediction, Feature selection, Deep learning, Large Language Models, Webserver (HTML, CSS, PHP)

Education

Ph.D. in Computational Biology	July 2021 – Present
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India	
M.Sc. in Biotechnology	2019
Ch. Charan Singh University (Campus), Meerut, India	
B.Sc. in Biotechnology	2017
Ch. Charan Singh University (DAV College), Meerut, India	

Research Experience

July 2021 -IIIT-Delhi, India

current *PhD Thesis:- Development of AI-based tools for designing subunit vaccines*
Advisor: Prof Gajendra P.S. Raghava

Dec.2020 - Premas Biotech Pvt. Ltd, Manesar, Gurugram (Haryana)

June 2021 *Project:- “AXTEX-4D” in Mammalian Cell Culture Lab Department*

Supervisor: Dr. Ambica Baru

Key Responsibilities: Cell line maintenance, data analysis, 3D culture, Spheroids formation by using hanging drop method.

Jan.2019 - Institute Of Nanoscience & Technology, Mohali (Punjab)

July 2019 *Project:- Peptide-Based Theranostics In Treating Glioblastomas*

Supervisor: Dr. Jiban Jyoti Panda (Scientist ‘E’)

Key Responsibilities: Setting up instruments and laboratory equipment used to conduct and monitor experiments such as Centrifuges, Spectrophotometer, and Zeta sizer Nano ZSP.

Publications

PEER-REVIEWED JOURNALS

- **HAIRPred: Prediction of Antibody-Interacting Residues**
R. Sahni, Nishant Kumar, Raghava GPS Protein Science, 2025
- **Prediction of Anti-Freezing Proteins From Their Evolutionary Profile**
Kumar N, Choudhury S, Bajiya N, Patiyal S, Raghava GPS Proteomics, 2024
- **A method for predicting linear and conformational B-cell epitopes in an antigen from its primary sequence**
Kumar N, Tripathi S, Sharma N, Patiyal S, Devi NL, Raghava GPS Computers in Biology and Medicine, 2024
- **Multi-perspectives and challenges in identifying B-cell epitopes**
Kumar N, Bajiya N, Patiyal S, Raghava GPS Protein Science, 2023
- **DMPPred: a tool for identification of antigenic regions responsible for inducing type 1 diabetes mellitus**
Kumar N, Patiyal S, Choudhury S, Tomer R, Dhall A, Raghava GPS Briefings in Bioinformatics, 2023
- **Prediction and design of thermostable proteins with a desired melting temperature**
Tijare P, Kumar N, Raghava GPS Scientific Reports, 2025

- **Prediction of inhibitory peptides against *E. coli* with desired MIC value**
Bajiya N, **Kumar N**, Raghava GPS Scientific Reports, 2025
- **Prediction of hemolytic peptides and their hemolytic concentration**
Rathore AS, **Kumar N**, Choudhury S, Mehta NK, Raghava GPS Communications Biology, 2025
- **Prediction of Plant Resistance Proteins Using Alignment-Based and Alignment-Free Approaches**
Gahlot PS, Choudhury S, Bajiya N, **Kumar N**, Raghava GPS Proteomics, 2025
- **HLA-DR4Pred2: An improved method for predicting HLA-DRB1*04:01 binders**
Patiyal S, Dhall A, **Kumar N**, Raghava GPS Methods, 2024
- **Dual Blood-Brain Barrier-Glioma Targeting Peptide-Poly(levodopamine) Hybrid Nanoplat-forms as Potential Near Infrared Phototheranostic Agents in Glioblastoma**
Dube T, **Kumar N**, Bishnoi M, Panda JJ Bioconjugate Chemistry, 2021

PRE-PRINTS

- **MAP format for representing chemical modifications, annotations, and mutations in protein sequences: An extension of the FASTA format**
Shendre A, Mehta NK, Rathore AS, **Kumar N**, Patiyal S, Raghava GPS arXiv (q-bio.BM), 2025
- **In silico tool for identification of colorectal cancer from cell-free DNA biomarkers**
Mathur K, Jain S, Bajiya N, **Kumar N**, Raghava GPS arXiv (q-bio.GN), 2025
- **PEPlife2: An updated repository of the half-life of peptides**
Alam U, Chaudhary K, **Kumar N**, Tomer R, Patiyal S, Raghava GPS bioRxiv, 2025

Teaching experience

Teaching Assistant, IIIT-Delhi, India

- Machine Learning in Biomedical Applications 2 Semesters (2022 & 2023)
- Big Data Mining in Healthcare 1 Semester (2023)

Key Responsibilities

- Prepared and deliver tutorials, graded assignments and provide constructive feedback to students
- Mentored bachelor's and master students on research projects

Conference and Achievements

1. **Presented - DMPPRed at JBI2025 XV Bioinformatics Symposium (JBI2025)** at Madrid, Spain, 22 – 24 October 2025
2. **Adipo-Cible Winter School — Obesity Unraveled: From Cellular Mechanisms to Therapeutic Pathways** held by Université Côte d'Azur, France, Dec. 16–19, 2024.
3. **InBix–ADNAT 2024 Conference:** Integrating Archaeogenetics, Forensics & Multi-omics for Human Health Breakthroughs, 28–30 November 2024, Banaras Hindu University, Varanasi, India.
4. **3-day Hands-on Workshop:** “Introductory Omics and RNA-seq Data Analysis,” 11–13 July 2024, organised by Bio-IT Centre, IBAB, Bengaluru, India.
5. **Workshop:** Establishing and Sustaining Cloud Infrastructure for Informatics-based Healthcare, 22–23 March 2024, IIIT-Delhi, India.
6. **CSIR-UGC JRF Fellowship (2020).**
7. **International Conference on GRISAAS** (Astha Foundation, Meerut): presented poster “Nano Fertilizers,” 28–30 October 2018.
8. **Member, Auditorium Committee — International Conference on Global Research Initiatives for Sustainable Agriculture & Allied Sciences (GIPRS-2018),** RARI Jaipur, 28–30 October 2018.
9. **International Conference on GRISAAS** (Astha Foundation, Meerut), 28–30 October 2018.

10. **International Workshop:** Advanced Biological Techniques and Intellectual Property Rights, 27–28 July 2018.
11. **Seminar cum Hands-on Workshop:** Molecular Biology and Bioinformatics, 8–10 December 2015.

Technical Skills

Programming: Python, R, Bash, Large Language Models

Hobbies: Book Reading, Social Service, Travelling, Sketching, Journaling

Hands-On Instrument: UV-Vis spectrophotometer, Zetasizer Nano ZSP, Bright Field Microscope, FACS (Fluorescence-activated cell sorting), Centrifuge Machine

Strengths: Excellent team leader with team building capability, creativity, sincerity, good judgment, determination, self development, flexibility, commitment, loyalty, Good motivator and disciplined.

Referees

Name Prof. Gajendra P.S. Raghava
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Tel +91-11-26907444
Address Director, IIIT-Delhi, India

Name Dr. Ambica Baru
Contact ambica.baru@premasbiotech.com
Addresse Premas Biotech, Manesar, Haryana, India
Scientist
Mammalian cell culture lab